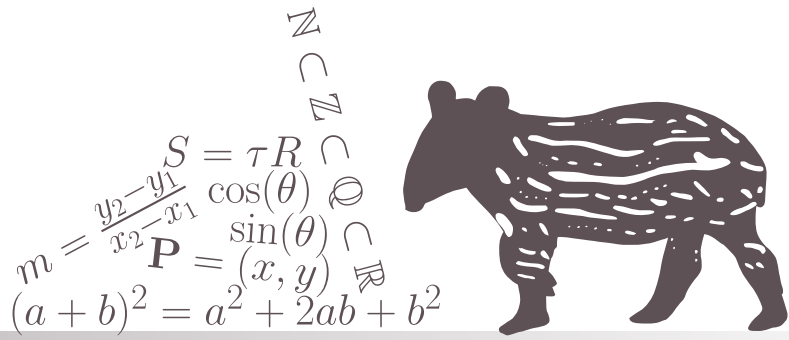


CHAPTER

1



MATHEMATICAL BACKGROUND

This is a test text for the **Mathematical background** chapter.

Is this what's written in the actuall chapter header?

It should be!

1.1 PREFACE

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Testing some equations: let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\mathbf{c} = \alpha \mathbf{x} + \beta \mathbf{y} \in \mathbb{R}^n. \quad (1.1)$$

However, if $\mathbf{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha \mathbf{x} \in \mathbb{R}^m$, and therefore $\mathbf{c}$ is not defined (i.e. [Equation 1.1](#) is irrelevant).

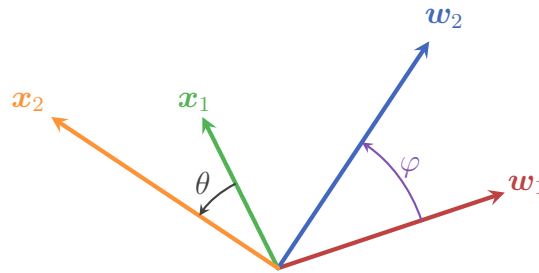


Figure 1.1: This is the first test figure. Does it look ok?

Note 1.1 Test note

This is a test note. I will try to also reference it later.



Now, I wish to reference [1.1](#). Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = [1, 2, -3]$ and $y = [5, -1]$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha x \in \mathbb{R}^3$, while $\beta y \in \mathbb{R}^2$. This means that $\alpha x + \beta y$ is not defined.



I believe it is now time for some figures! See [Figure 1.1](#).

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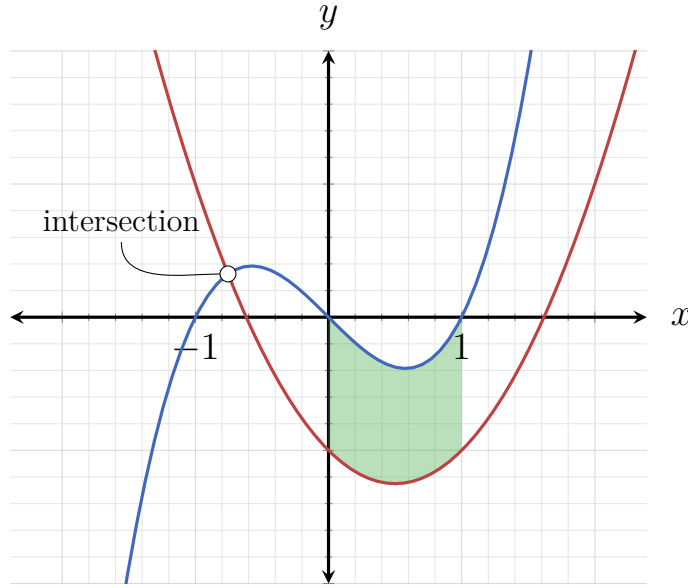


Figure 1.2: And this is another test figure! I hope it works.

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1.1. PREFACE

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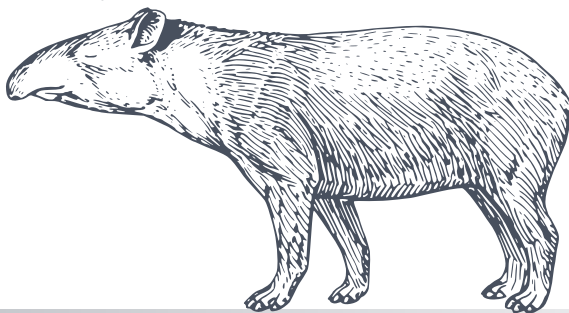
$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i \quad \vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \quad \mathbf{A} \vec{x} = \lambda \vec{x} \quad (\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n \quad \det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_j^i = \langle a^i | b_j \rangle$$



CHAPTER

2

THE BASICS OF LINEAR ALGEBRA

This chapter introduces the most important concepts of linear algebra, such as vector spaces, linear transformations, and eigenvectors and eigenvalues. Starting with simple 2- and 3-dimensional geometric concepts, the chapter guides the reader through the topics by introducing them in a visual and intuitive way, only generalizing them afterwards.

After reading this chapter and successfully completing the assignments at the end, the reader should have an intuitive understanding of basic linear algebra.

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

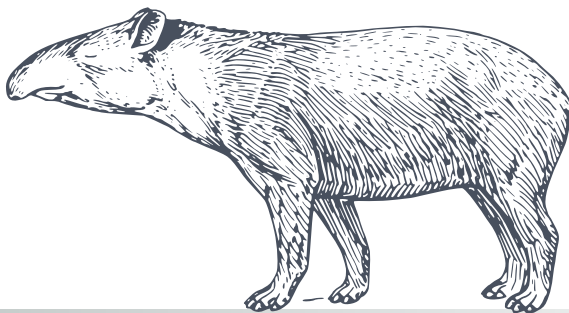
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

3

INTRODUCING SOME FORMALISM

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

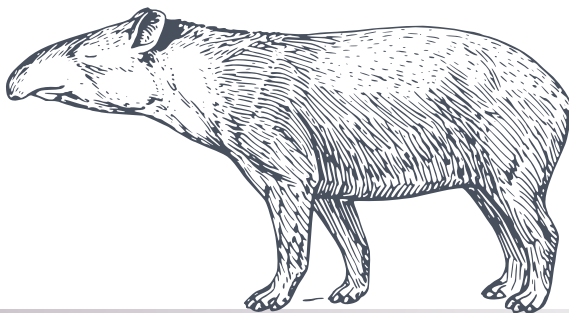
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

4

ADVANCED TOPICS IN LINEAR ALGEBRA

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

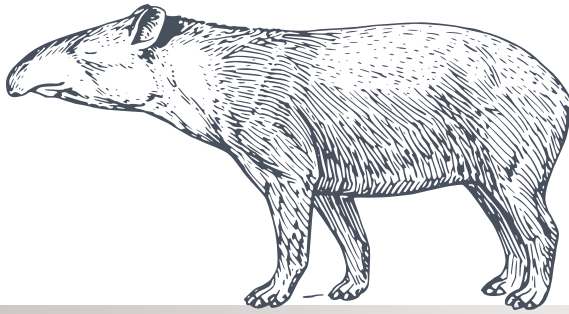
$$(\vec{A} \cdot \vec{B})^T = \vec{B}^T \cdot \vec{A}^T$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

5

FURTHER RELATED TOPICS

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

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CHAPTER 5. FURTHER RELATED TOPICS

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